

Department of Mathematics and Natural Sciences
MAT110 : Differential Equations and Coordinate Geometry
Assignment-01 and 02
Deadline: 30/03/2026

Summer 2026

Total Marks: 15+15

(FROM LIMITS:)

1. Find the limit of the following functions:

[1+1+1]

- a. $\lim_{x \rightarrow +\infty} (1 + a/x)^{bx}$
- b. $\lim_{x \rightarrow 1} (2 - x)^{\tan[(\pi/2)x]}$
- c. $\lim_{t \rightarrow -2} \frac{t^3 + 8}{t + 2}$

2. $f(x) = \begin{cases} \frac{1}{x+2} & , x < -2 \\ x^2 - 5 & , -2 < x < 3 \\ \sqrt{x+13} & , x > 3 \end{cases}$; find $\lim_{x \rightarrow -2} f(x)$ and $\lim_{x \rightarrow 3} f(x)$ [2]

(FROM CONTINUITY)

3. Let $f(x) = |x| + |x - 1|$. Test continuity of $f(x)$ at $x = 0$. [3]

4. $f(x) = \begin{cases} x^3 \sin(1/x) & , x \neq 0 \\ 0 & , x = 0 \end{cases}$ at $x = 0$ [3]

Is $f(x)$ continuous at $x = 0$? Show mathematically.

(FROM DIFFERENTIABILITY)

5. Let $f(x) = \begin{cases} x^2 - 16x, & x < 9 \\ 12\sqrt{x}, & x \geq 9 \end{cases}$. Determine whether $f(x)$ is differentiable at $x = 9$. [2]

6. Test the differentiability of the following function,

$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$ at $x = 0$ [2]

7. Differentiate $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ with respect to $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$. [2]

(FROM INCREASING AND DECREASING FUNCTION:)

8. Find (a) the intervals on which f is increasing, (b) the intervals on which f is decreasing, (c) draw the graph of the functions. [2+2+2]

(i) $f(x) = 3x^5 - 5x^3$

(ii) $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$

(iii) $f(x) = x^3 + 3x^2 - 9x + 1$

9. Find the relative extrema (maxima/ minima) using both the first and second derivative tests. Show the results by drawing the graphs of the functions. [2.5+2.5]

(i) $f(x) = 2x^3 - 9x^2 + 12x$

(ii) $f(x) = \frac{x}{2} - \sin x, \quad 0 < x < 2\pi.$

(Taylor series expansion:)

10. Calculate the Taylor series expansion of $f(x) = \sin(x)$ at $x = \pi/2$.

[2]